

Review

Leveraging Municipal Solid Waste Management with Plasma Pyrolysis and IoT: Strategies for Energy Byproducts and Resource Recovery

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Abstract: The escalating challenges of municipal solid waste (MSW) management, exacerbated by the classification of MSW as hazardous waste due to the presence of heavy metals (HMs) and toxic compounds, necessitate innovative treatment strategies. Plasma pyrolysis has emerged as a promising technology for converting MSW into valuable energy byproducts, such as syngas, bio-oil, and slag, while significantly reducing waste volume. However, maintaining optimal operational parameters during the plasma pyrolysis process remains a complex challenge that can adversely affect both the efficiency and the quality and quantity of outputs. To address this issue, the integration of the Internet of Things (IoT) presents a transformative approach. By leveraging IoT technologies, real-time monitoring and advanced data analytics can be employed to optimize the operational conditions of plasma pyrolysis systems, ensuring consistent performance and maximizing resource recovery. This review explores the synergistic integration of plasma pyrolysis and IoT as a novel strategy for MSW management. The slag from plasma treatment can be efficiently channeled into anaerobic digestion (AD) systems, promoting resource recovery through biogas production and the generation of nutrient-rich digestate. This synergy not only mitigates the environmental impacts associated with traditional MSW disposal methods but also paves the way for sustainable energy recovery and resource management. Ultimately, this review presents a comprehensive framework for exploiting plasma pyrolysis and IoT in addressing the pressing issues of hazardous MSW, thereby fostering a circular economy through innovative waste-to-energy solutions.

Keywords: municipal solid waste; plasma pyrolysis; internet of things; anaerobic digestion; process optimization; sustainable waste treatment



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1. Introduction

The management of municipal solid waste (MSW) has become one of the most critical environmental challenges facing urban areas worldwide. In 2016, the global generation of MSW reached approximately 2.01 billion metric tons, a figure projected to surge to 3.4

billion metric tons by 2050 [1]. This alarming trend is primarily driven by rapid urbanization, population growth, and changing consumption patterns, leading to increased waste generation and disposal challenges. A significant concern is that a substantial portion of this waste is classified as hazardous due to the presence of heavy metals (HMs), toxic chemicals, and organic pollutants [2]. According to Ashutosh et al. [3], roughly 15% of MSW contains hazardous materials, highlighting an urgent need for effective treatment and management strategies that can mitigate the associated risks. The hazardous nature of MSW is underscored by its diverse composition, which includes plastics, metals, glass, organic matter, and electronic waste (e-waste) [4]. HMs such as lead, mercury, cadmium, and arsenic are particularly worrisome, as they can leach into soil and groundwater, leading to environmental contamination and adverse health effects [5]. For instance, lead exposure is linked to severe neurological damage, especially in children, while mercury is known to impact the nervous system and kidneys [6,7]. The improper disposal of e-waste further exacerbates the issue, as it often contains high concentrations of these hazardous substances. The Global E-waste Monitor 2020 reported that 53.6 million metric tons of e-waste were generated in 2019, with only 17.4% being recycled properly [8]. This improper disposal not only leads to the release of harmful substances into the environment but also represents a significant loss of valuable materials. Therefore, addressing the hazardous components of MSW is critical for both environmental protection and resource recovery.

To tackle the complex challenges posed by hazardous MSW, innovative treatment technologies are essential. Plasma pyrolysis is one such advanced thermal treatment technology that utilizes high-temperature plasma to decompose organic materials in waste [9]. Operating at temperatures exceeding 1000 °C, plasma pyrolysis effectively breaks down complex organic compounds into simpler molecules, including syngas, a mixture of hydrogen and carbon monoxide that can be harnessed for energy production [10]. Unlike conventional incineration and gasification, plasma pyrolysis minimizes the release of harmful emissions and allows for the effective treatment of hazardous waste components [9]. Fasihi et al.'s [11] study indicates that plasma pyrolysis can achieve waste volume reductions of up to 90%, transforming hazardous MSW into valuable energy byproducts while safely immobilizing HMs in a solid residue. The solid residue produced from plasma pyrolysis is characterized by its stable mineral composition, making it suitable for further processing through anaerobic digestion (AD). This biological treatment method converts organic matter into biogas and nutrient-rich digestate, thereby enhancing resource recovery [12]. The biogas produced can serve as a renewable energy source, primarily composed of methane and carbon dioxide. According to the International Renewable Energy Agency, biogas production from organic waste has the potential to provide a sustainable energy solution, with estimates suggesting that approximately 1000 terawatt-hours of electricity could be generated annually worldwide from biogas [13]. Furthermore, the digestate can be utilized as a fertilizer, promoting sustainable agricultural practices and closing the nutrient loop.

The data collection procedures for this review were conducted in English by searching various scientific citation indexing services, including Web of Science (<https://www.webofscience.com/>, accessed on 19 August 2024), Scopus, Google Scholar, and other online resources. The search keywords used were: “Sustainable waste treatment with plasma pyrolysis”, “Sustainable MSW management approach”, “Integrating the IoT with solid waste management”, and “Techniques to optimize resource recovery and energy materials from municipal solid waste”. Several hundred articles were initially retrieved and then refined to focus on specific topics related to biochar, pyrolysis, plasma pyrolysis, AD, resource recovery from MSW, the IoT, optimization processes, energy byproducts, and integrated technologies. Ultimately, 71 published papers that were closely relevant to the scope of this study were selected for inclusion in this review.

Integrating the IoT into MSW management systems presents a significant opportunity to optimize the entire waste treatment process. IoT technologies enable real-time monitoring of waste generation, collection, and processing, facilitating data-driven decision-making. Based on Sahar et al. [14], by deploying sensors and smart bins, municipalities can track waste levels and optimize collection routes, reducing operational costs and emissions associated with waste transportation. Moreover, IoT can enhance the efficiency of thermal processes and AD systems by providing real-time data on temperature, pressure, and feedstock composition [15,16].

The growing challenges posed by MSW, particularly its hazardous components, require innovative treatment strategies. This review recommends using IoT to enable precise control over processes, ensuring optimal performance and resource recovery. Plasma pyrolysis, integrated with IoT technologies, will offer a promising solution for sustainable MSW management. It can convert MSW into valuable energy byproducts and nutrient-rich resources, addressing environmental concerns and supporting a circular bio-economy. The byproducts, such as syngas and biochar, can generate heat and electricity, while slag or biochar can enhance AD by closing the loop, improving biogas production and resource recovery. Overall, these IoT-enabled applications can optimize energy byproduct production and resource utilization.

2. Internet of Things Overview

The IoT represents a transformative paradigm that connects physical devices to the Internet, enabling them to collect, exchange, and analyze data [17]. This interconnected network has profound implications across various sectors, including healthcare [18], agriculture [19], transportation [20], and smart cities [21]. The foundation of IoT lies in its ability to integrate sensors, software, and other technologies into everyday objects, facilitating real-time data acquisition and communication. At the core of IoT systems are sensors that gather data from the environment. These can include temperature, humidity, motion, and gas sensors, among others. Mehar et al. [15] investigated the industrial-scale pyrolysis process for recycling carbon fiber-reinforced polymer (CFRP) composite waste through the integration of IoT technology. Their study recommends the use of sensors and actuators to gather data from the recycling facility, considering that the incoming composite waste will exhibit diverse types and properties. Consequently, the suggested IoT framework was designed to adjust the recycling process parameters based on the specific composites, ensuring a standardized pyrolysis setup. They concluded that implementing this IoT framework would enhance the adaptability of the recycling process across different CFRP waste types, ultimately leading to more efficient energy utilization, scalability, increased reliability, and a reduction in both cost and treatment time. In smart AD systems, the reactor's performance was monitored remotely through the installation of temperature, pH, and oxidation-reduction potential electrodes linked to a Programmable Logic Controller (PLC). Additionally, the quality of the biogas produced was tracked using an online biogas analyzer, allowing for real-time monitoring. This setup enabled centralized operators to enhance process performance, leading to improved operation and maintenance of decentralized AD systems [16].

Based on Cüneyt et al. [22], data collected from these sensors is transmitted through various communication protocols, such as Message Queuing Telemetry Transport (MQTT) and Constrained Application Protocol (CoAP), and WebSocket protocols, which are designed for low-bandwidth, high-latency environments. Once the data reaches a centralized system or cloud platform, it undergoes processing using advanced algorithms. Machine learning (ML) and artificial intelligence (AI) play pivotal roles in analyzing these data [23]. For example, algorithms like Random Forest (RF), Support Vector Machines (SVM), and

Neural Networks can be utilized to identify patterns and make predictions based on historical data. Cuong et al. [24] conducted a study comparing six ML models to forecast MSW generation in selected residential areas of Vietnam. Among these models, the k-nearest neighbor and RF algorithms demonstrated strong predictive performance on the training dataset (which comprised 80% of the total data), achieving an R^2 value exceeding 0.96 and a mean absolute error ranging from 121.5 to 125.0 on the testing dataset (the remaining 20%). The ML models developed in their study offer dependable predictions regarding MSW generation, which will be instrumental in the design, planning, and execution of an integrated MSW management strategy in the country. In addition, Johanna et al. [25] presented a study that involved the application and evaluation of two AI techniques for addressing MSW issues. They utilized long short-term memory (LSTM) and SVM networks. The LSTM implementation considered various configurations, annual assessments, and temporal filtering of MSW collection intervals. The findings indicated that the SVM approach effectively adapted to the chosen data, producing reliable regression curves even with minimal training data, resulting in more precise outcomes compared to those achieved with the LSTM technique. Figure 1 shows basic architecture of IoT.

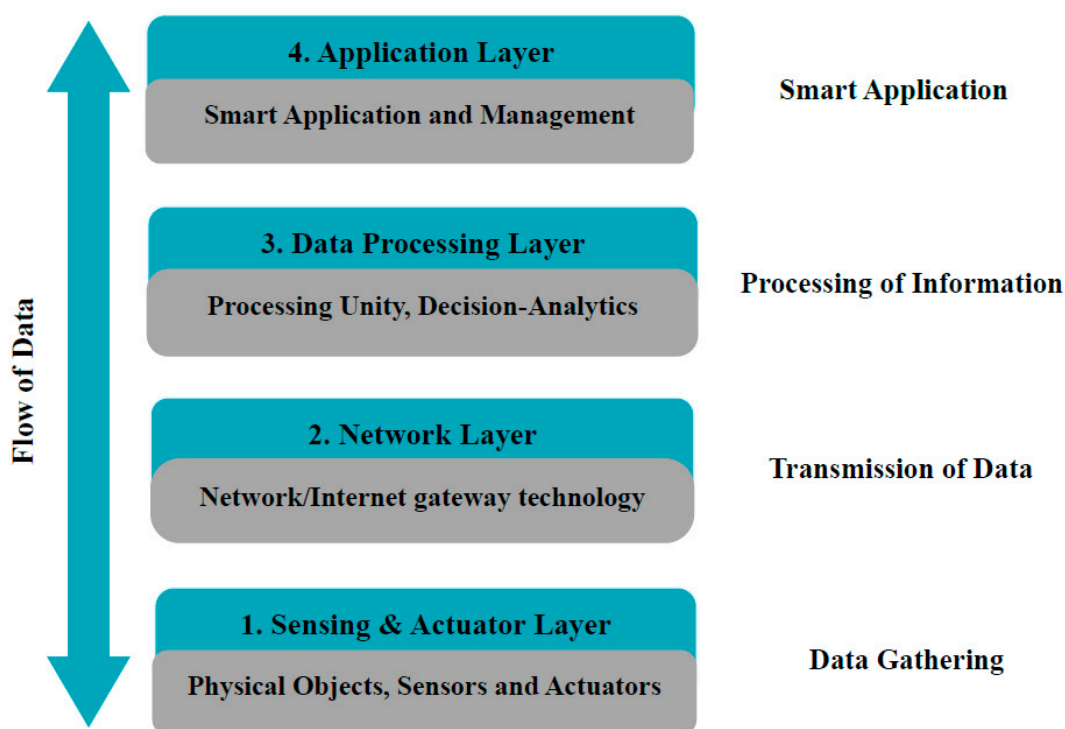


Figure 1. Architecture of the internet of things.

In addition, Abdullah et al. [26] introduced a comprehensive MSW management framework that utilizes cloud analytics, algorithms, and IoT to enhance the efficiency and sustainability of waste collection in smart cities. Their main contributions include: (1) a prototype of an ultrasonic sensor designed for accurate monitoring of bin fill levels; (2) a networking architecture using LoRaWAN and cellular technology to ensure city-wide connectivity; (3) a cloud-based system for collecting, storing, and analyzing large volumes of sensor data; and (4) optimization algorithms aimed at reducing collection routes and fuel consumption based on real-time data from waste bins. The outcomes demonstrated an 18% decrease in vehicle maintenance costs compared to traditional methods, a 29% reduction in fuel usage and emissions, a 33% rise in waste processing capacity, and a 32% increase in route efficiency. In a study conducted by Sahar et al. [14] an IoT-based solution was introduced to enhance waste-management processes in Islamabad, Pakistan. The researchers

developed an algorithmic method that enables dumpsters to communicate their waste levels to a municipal authority server through a wireless network before collection begins. The findings indicate significant improvements in both the time required for waste collection and the distance covered, thanks to the implementation of a heuristic algorithm. In traditional waste collection methods, the average time taken was 5.6 h with a total distance of 344.3 km (with the best recorded time at 4.99 h and distance at 339.84 km). In contrast, the smart dumpster approach reduced the average time to 4.365 h and the distance to 234.7 km (with the best results being 3.06 h and 226.2 km). Furthermore, Amira et al. [27] reported a groundbreaking IoT-based smart MSW management system that incorporates a variety of advanced technologies. This system features a smart modular garbage bin, capabilities for route selection and optimization, and real-time notifications for all involved parties. By utilizing a global positioning system, fire, temperature, ultrasonic sensor, and LED indicators within the smart bins, the system effectively tracks waste levels and facilitates timely collection, therefore minimizing the risk of overflow and potential ecological issues. Additionally, the integration of LoRa communication technology and the MQTT protocol greatly enhances the system's reliability and overall efficiency.

3. Assessment of Plasma Pyrolysis Technology

Plasma pyrolysis is an advanced waste treatment method that leverages the unique properties of plasma to break down complex hydrocarbons into simpler molecules [10]. For over four decades, thermal plasma processes have demonstrated significant technological advantages across various applications. Trelles et al., [28] the appeal of thermal plasmas lies in their high energy density, which ranges from 10^6 to 10^7 J/m³, accompanied by impressive heat flux densities of about 10^7 to 10^9 W/m². Additionally, these processes feature rapid quenching rates between 10^6 and 10^8 K/s and high processing speeds. Generally, direct current (DC) arc plasma torches serve as the core component in these applications, which encompass plasma spraying, ultra-fine particle synthesis, metal welding, and cutting, as well as extractive metallurgy, waste treatment, and bio-energy production [29]. These torches function as electrical, chemical, and thermal devices, enabling material modifications that are often unattainable or economically impractical with alternative technologies. Moreover, plasma spraying, which utilizes DC arc plasma torches, serves as a notable application that has gained significant traction across multiple fields, including waste treatment. For example, Kangana et al. [9] highlighted this method as one of the most environmentally friendly and cost-effective approaches for managing solid waste. As, plasma pyrolysis effectively immobilizes hazardous materials, including HMs, within a stable glassy residue, reducing their leachability and environmental impact. The simple treatment of MSW through plasma pyrolysis is presented in Figure 2.

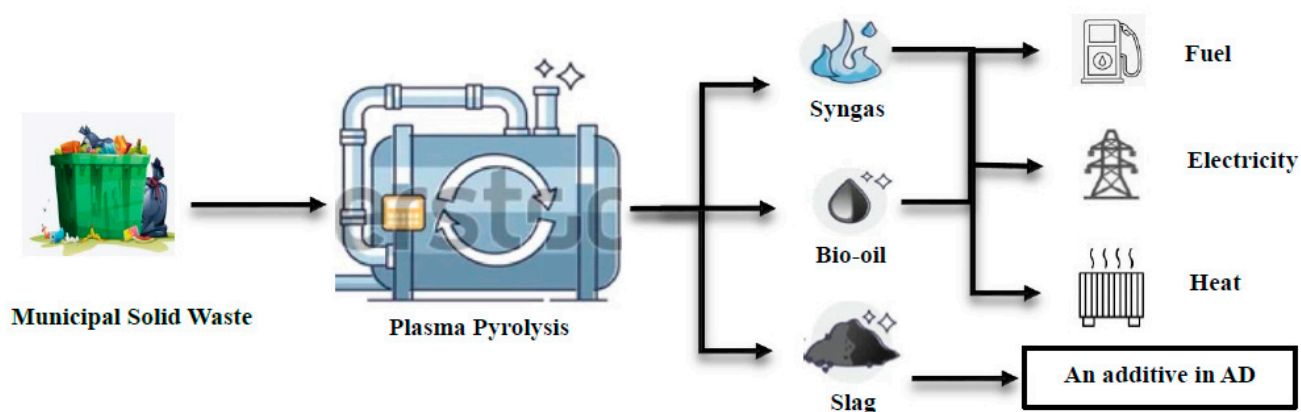


Figure 2. Diagram of municipal solid waste treatment under plasma pyrolysis.

Figure 1, depicting MSW treatment through plasma pyrolysis, illustrates a transformative process that enhances waste management while maximizing resource recovery. Plasma pyrolysis operates at temperatures ranging from 1000 °C to 5000 °C, effectively decomposing organic and inorganic materials into valuable byproducts such as syngas, liquid fuels, and solid slag in an oxygen-free environment [30]. Moreover, the solid slag generated from the process is noteworthy; it can be repurposed as an additive in AD, enhancing the biodegradability of organic matter and improving biogas yields [12]. In Idris et al. [31] study, various single plastics commonly found in high concentrations within MSW were examined. These included high-density polyethylene (HDPE), low-density polyethylene (LDPE), polypropylene (PP), polystyrene (PS), and polyethylene terephthalate (PET). The study utilized a two-stage pyrolysis process in a non-thermal plasma/catalyst reactor to evaluate the production of hydrogen and other gases. Additionally, a blend of these five plastics was created in proportions reflective of those typically present in MSW, simulating a waste plastic composition with known characteristics. The research also incorporated “real-world” mixed plastics sourced from various commercial and industrial origins into the reactor system to facilitate hydrogen production. The author concluded that the hydrogen yield was significantly higher in thermal plasma compared to non-thermal plasma when processing MSW. In addition, Messerle et al. [32] generated 88.3% of hydrogen from plasma pyrolysis of propane–butane gas mixture. According to Vyas et al. [33] plasma pyrolysis is an environmentally friendly waste treatment method that is recognized as a strategy to mitigate global warming under the Kyoto Protocol. This process adheres to a zero-discharge philosophy, ensuring that all input waste substances are recycled back into nature or the market in a way that safeguards the ecosystem. Plasma technology is a versatile solution capable of processing various waste types while generating hydrogen, electricity, and heat without releasing harmful substances such as dioxins, furans, etc. Furthermore, MSW is considered hazardous due to the presence of various waste substances, including e-waste [4]. Chiang et al. [34] have reported plasma pyrolysis as a sustainable treatment method for e-waste materials as it allows the destruction of toxic chemical substances environmentally and economically. Therefore, the plasma pyrolysis approach not only addresses the challenges of MSW management but also promotes sustainability by converting waste into valuable resources, positioning it as a superior alternative to conventional treatment processes. Table 1 shows hydrogen production from different waste materials processed under plasma pyrolysis.

Table 1. Current Applications of Plasma Pyrolysis in Waste Management.

Waste Types	Process Overview	Hydrogen Gaseous Product (%)	Reference
Propane–butane gas mixture	The author investigated both thermodynamic and experimental analyses of a mixture of hydrocarbon gases under atmospheric pressure within an electric arc plasma reactor.	88.3	[32]
Petrochemical spent catalyst	The DC arc plasma technology was utilized with argon as the carrier gas	53.20	[35]
Medical waste	Nitrogen served as the carrier gas in the DC plasma arc method, with a power input of 50,000 W.	40.83	[36]
Hydrocarbons	Electric arc plasma was used as the heat source, with argon serving as the carrier gas	42.5	[37]

Table 1. Cont.

Waste Types	Process Overview	Hydrogen Gaseous Product (%)	Reference
Electronic waste	Argon was used as the carrier gas in alternating current arc plasma technology.	36.18	[38]
High-density polyethylene	Various types of single and mixed plastics, commonly found in high concentrations in MSW, were examined using electric arc plasma pyrolysis with nitrogen and argon as carrier gases.	65.9	[31]
Low-density polyethylene		61.3	
Polystyrene		67.8	
Polyethylene terephthalate		24.3	
Polypropylene		53.1	
Tire Waste	The DC arc plasma method was utilized, using argon as the carrier gas.	54.69	[39]
Municipal Solid Waste		43.50	

4. Environmental Pollution from Traditional Treatment of Municipal Solid Waste

Traditional treatment methods for MSW, primarily landfilling and incineration, significantly contribute to environmental pollution, posing serious risks to human health and ecosystems. Global MSW generation is projected to increase significantly by 2050, nearly doubling the current figure of 2 billion tons per year, with much of this waste ending up in landfills or being incinerated [1]. Landfills, which currently account for approximately 95% of waste disposal worldwide, produce leachate, a toxic liquid formed when rainwater filters through waste, which can contaminate groundwater and surface water. Studies have shown that leachate from landfills contains harmful substances such as HMs and organic pollutants, posing significant health risks to communities nearby. For instance, Prasanna et al. [40] reported that leachate from landfills can lead to elevated levels of contaminants like benzene and toluene in groundwater. Furthermore, landfills are a major source of methane emissions, a potent greenhouse gas with a global warming potential 25 times greater than carbon dioxide [41]. The U.S. Environmental Protection Agency (EPA) reports that landfills account for about 15% of total methane emissions in the country, contributing significantly to climate change [42]. On the other hand, incineration, while reducing waste volume, releases harmful pollutants such as dioxins, furans, and particulate matter into the atmosphere [43]. Exposure to these pollutants is linked to serious health issues, including respiratory diseases and cancer. Communities near incinerators often experience higher rates of health problems; for example, Jose et al. [44] indicated that individuals living close to waste incineration plants had a 20% higher risk of developing respiratory illnesses. The socioeconomic implications are also significant, as low-income communities disproportionately bear the burden of these environmental hazards, leading to increased healthcare costs and diminished quality of life.

In addition, recycling is often promoted as a more environmentally friendly alternative to landfilling. It conserves resources, reduces energy consumption, and minimizes pollution. According to the EPA [45], recycling and composting prevented the release of approximately 186 million metric tons of carbon dioxide equivalent into the air in 2018. However, recycling is not without its challenges. Contamination of recyclable materials can lead to increased costs and reduced efficiency in recycling processes. For instance, Akan et al. [46] indicated that contamination rates in curbside recycling programs can range from 15% to 25%, making it difficult to process materials effectively. Furthermore, while recycling reduces the volume of waste sent to landfills, it does not eliminate the environmental impact. Thus, while recycling is a crucial component of waste management, it must be implemented alongside other strategies to maximize its benefits. Moreover, composting is another sustainable waste-management practice that can significantly reduce the envi-

ronmental impact of organic waste. By diverting MSW from landfills, composting reduces methane emissions and produces valuable soil amendments. Based on EPA [45], composting can reduce greenhouse gas emissions by up to 50% compared to landfilling organic waste. However, the effectiveness of composting is contingent upon proper management. If not executed correctly, composting can lead to the release of odors and attract pests, which can pose public health concerns [47]. Additionally, there are challenges associated with contamination in composting. If non-biodegradable materials, such as plastics, are included in the composting process, they can contaminate the resulting compost, making it unsuitable for agricultural use. Costanza et al. [48] found that compost from contaminated sources often contained microplastics, which can have detrimental effects on soil health and crop production. Therefore, while composting is a valuable tool for managing organic waste, it requires careful oversight to ensure its effectiveness and safety.

Furthermore, AD is increasingly recognized as a viable alternative for managing organic waste, converting it into biogas and digestate. However, the digestate produced can be a source of HM contamination if the input materials are not carefully selected. Fernanda et al. [49] found that digestate from AD can contain significant levels of HMs such as zinc, copper, and lead, especially when sourced from contaminated organic waste. This poses risks when the digestate is used as a fertilizer, potentially leading to soil and water contamination. In addition, the presence of those HMs may lead to the instabilization of the AD process, which leads to a decrease in biogas yield [50]. If not properly managed, the application of contaminated digestate can lead to the accumulation of HMs in agricultural soils, adversely affecting crop quality and human health. Moreover, gasification and pyrolysis are thermal technologies that aim to convert waste into energy while minimizing environmental impacts. These processes involve heating waste in a limited amount of oxygen and the absence of oxygen, respectively, to produce syngas, bio-oil, and biochar, which can be used for energy production. While they present potential benefits over traditional incineration, concerns remain regarding the emissions of toxic substances and the overall efficiency of these technologies. Yong et al. [51] showed that gasification can produce harmful byproducts, including volatile organic compounds and particulate matter, which can contribute to air pollution if not properly managed. Furthermore, the environmental benefits of gasification and pyrolysis depend heavily on the quality of the feedstock and the efficiency of the technology used. For instance, byproducts from this process may contain HMs and other toxic chemical compound [51,52]. Therefore, the conventional treatment of MSW through the aforementioned technologies poses significant environmental pollution challenges. From groundwater contamination and methane emissions to toxic gas emissions and HM accumulation, the impacts are far-reaching and often disproportionately affect vulnerable communities. As global MSW generation continues to rise, it is imperative to adopt comprehensive waste-management strategies that prioritize environmental protection, bio-energy production, and resource recovery, ultimately leading to a more sustainable future. Table 2 presents process overview, merits and demerits of different treatment methods for MSW.

Table 2. Process Overview, Merits and Demerits of Different Treatment Methods for Municipal Solid Waste.

Treatment Methods	Process Overview	Merits	Demerits	References
Open Dumping	Disposal of waste in open areas without any environmental controls	Low cost and easy to implement	Causes severe environmental pollution and health risks	[53]

Table 2. *Cont.*

Treatment Methods	Process Overview	Merits	Demerits	References
Sanitary Landfilling	Controlled disposal of waste in designated sites, with measures to protect the environment	Reduces environmental impact compared to open dumping; can capture landfill gas for energy	Land use issues; potential for groundwater contamination	[54]
Recycling	Process of collecting and processing materials to create new products	Conserves natural resources; reduces landfill waste; saves energy	Contamination of recyclables can hinder processes; only limited type of waste can be recycled	[55]
Vermiculture	Use of earthworms to decompose organic waste into nutrient-rich compost	Produces high-quality compost; reduces organic waste volume	Requires careful management, not suitable for all waste types, produces toxic gas	[56]
Composting	Biological decomposition of organic waste into compost under controlled conditions	Enhances soil quality, reduces landfill waste, carbon sequestration	Can produce odors; requires space and management, not suitable for all waste types	[47]
Anaerobic Digestion	Breakdown of organic matter in the absence of oxygen, producing biogas and digestate	Generates renewable energy, promotes nutrient recovery	Requires careful management, cannot immobilize heavy metals, which leads to the second pollution via further use of digestates, only organic materials can be treated	[49]
Incineration	Combustion of waste at high temperatures to reduce volume and generate energy	Reduces waste volume significantly, can generate electricity	Produces harmful emissions and ash, high operational costs	[43]
Gasification	Conversion of organic materials into syngas through partial oxidation	Produces cleaner energy than incineration, reduces waste volume	Can release pollutants, complex technology	[51]
Plasma gasification	Uses plasma arcs to convert waste into syngas and vitrified ash at high temperatures	High efficiency, reduces hazardous waste, produces valuable byproducts	High energy consumption, expensive technology, toxic compounds may present in byproducts	[57]
Pyrolysis	Thermal decomposition of organic materials in the absence of oxygen, producing oil, gas, and char	Reduces waste volume, recovers energy, versatile feedstock	Can produce toxic byproducts, cannot fix heavy metals, significant energy input.	[52]
Plasma Pyrolysis	Advanced pyrolysis using plasma technology to decompose waste	High efficiency and low emissions, can handle a wide range of waste types, immobilize heavy metals	Still in developmental stages; high costs and energy requirements	[10]

5. Leveraging Municipal Solid Waste Management with Plasma Pyrolysis and Internet of Things

Improving the management and treatment of MSW by integrating plasma pyrolysis with the IoT provides an innovative solution for waste disposal. Plasma pyrolysis is a high-temperature process that thermally decomposes organic materials in the absence of oxygen, converting waste into valuable byproducts such as syngas, bio-oil, which can be used for energy production, and slag, which has numerous industrial applications [10,58]. Integrating IoT technologies into this process allows for real-time monitoring and optimization of operational parameters, leading to improved efficiency and reduced emissions. For instance, IoT sensors can track temperature, pressure, and feedstock composition, enabling operators to make data-driven decisions that enhance the pyrolysis process [15]. According to Beni

et al.'s study [59], integrating the IoT into waste-management systems can enhance collection routes, achieving efficiency gains of 15–23% and boosting recycling rates by 10–17% while also cutting operational costs by USD 9000 to USD 13,000 monthly. Additionally, AI-driven algorithms have improved sorting precision and recycling rates, especially in areas with varied waste types, resulting in an 18% increase in efficiency and up to a 20% rise in recycling rates. Furthermore, the use of big data analytics has enabled more informed decision-making and strategic planning, leading to a 15–20% improvement in efficiency and a 12–17% increase in recycling rates. Additionally, the ability to remotely monitor and control the pyrolysis units can lead to significant reductions in downtime and maintenance costs. Bingbing et al., [60] reported that, implementing AI in waste logistics can lead to a reduction in transportation distances by as much as 36.8%, resulting in time savings of around 28.22% and cost savings of up to 13.35%. AI enhances the sorting and identification of waste, achieving accuracy levels between 72.8% and 99.95%. Moreover, the integration of chemical analysis with AI enhances the pyrolysis of waste, optimizes energy conversion processes, and improves carbon emission assessments. By leveraging IoT, the plasma pyrolysis process not only becomes more efficient but also contributes to a sustainable circular economy, turning waste into a resource while minimizing environmental impact.

However, MSW is characterized by its diverse composition, which poses significant challenges for conventional treatment methods [4]. This complexity makes plasma pyrolysis an ideal solution, as it is one of the most sustainable options for managing MSW [9]. Plasma pyrolysis can effectively process both organic and inorganic materials, including electronic waste [34], by breaking them down at high temperatures in an oxygen-free environment. This capability not only enhances waste treatment efficiency but also transforms various waste types into valuable byproducts, making plasma pyrolysis a versatile and environmentally friendly approach to MSW management [30]. While there may not yet be a comprehensive study that specifically integrates the IoT with plasma pyrolysis, the potential benefits of combining these technologies can be inferred from existing research on IoT applications in traditional pyrolysis processes or with other treatment methods. The integration of IoT enhances operational efficiency by enabling real-time monitoring and data analytics, which can optimize the treatment process for maximum energy recovery and minimal emissions. For instance, Cheng et al. [61] created models to forecast the outcomes of both non-catalytic and continuous pyrolysis processes for waste plastics using ML algorithms. They compared four algorithms, such as the Gaussian process, SVM, artificial neural network (ANN), and decision tree (DT), along with various input datasets to identify the most effective input variables for achieving the highest accuracy. Among these methods, the DT model demonstrated impressive accuracy ($R^2 > 0.99$) and notable generalizability when tested with training data.

Moreover, Kai et al. [62] developed ML models based on ANN to predict the kinetics of biomass pyrolysis. They utilized datasets from properties of feedstock and thermogravimetric analysis across a variety of biomass sources to construct and evaluate the networks. The findings indicated that increasing the number of neurons within the network enhances prediction accuracy, with an optimal range of 7 to 11 neurons identified. Furthermore, the application of the particle swarm optimization algorithm markedly increased the prediction accuracy of the biomass pyrolysis-ANN model, reducing the largest deviation in activation energy prediction from 12.85% to 6.72%. Asya et al. [63] utilized Shapley Additive Explanations as an ML model to analyze the pyrolysis of biomass. The authors found that pyrolysis temperature is the most critical factor affecting the yields of byproducts. Parisa et al. [64] explored the application of ML methods in the gasification of face mask waste. They developed various polynomial regression algorithms to analyze the cold gas efficiencies, exergy, and lower heating values of syngas as well as emission levels and assessed their perfor-

mance. The results demonstrated the remarkable accuracy of the ML models in predicting the response variables, with many cases achieving R^2 value exceeding 98%, underscoring their effectiveness. In particular, the algorithms attained outstanding R^2 values of 99.33% for emissions, 98.93% for cold gas efficiency, and 99.85% for exergy. These remarkable findings validate the reliability and precision of these algorithms in delivering accurate predictions. Therefore, by employing IoT sensors, operators can gather critical data on temperature, pressure, and feedstock characteristics, allowing for precise adjustments that improve the overall effectiveness of the plasma pyrolysis process. Moreover, this synergy can facilitate predictive maintenance, reducing downtime and operational costs. Given the proven advantages of IoT in enhancing aforementioned systems, the application of IoT in plasma pyrolysis is likely to yield significant improvements in waste-management efficiency, sustainability, and resource recovery, making it a promising area for future research and development. Figure 3 shows a schematic diagram of MSW treatment using plasma pyrolysis and co-processing with AD, integrated with the IoT.

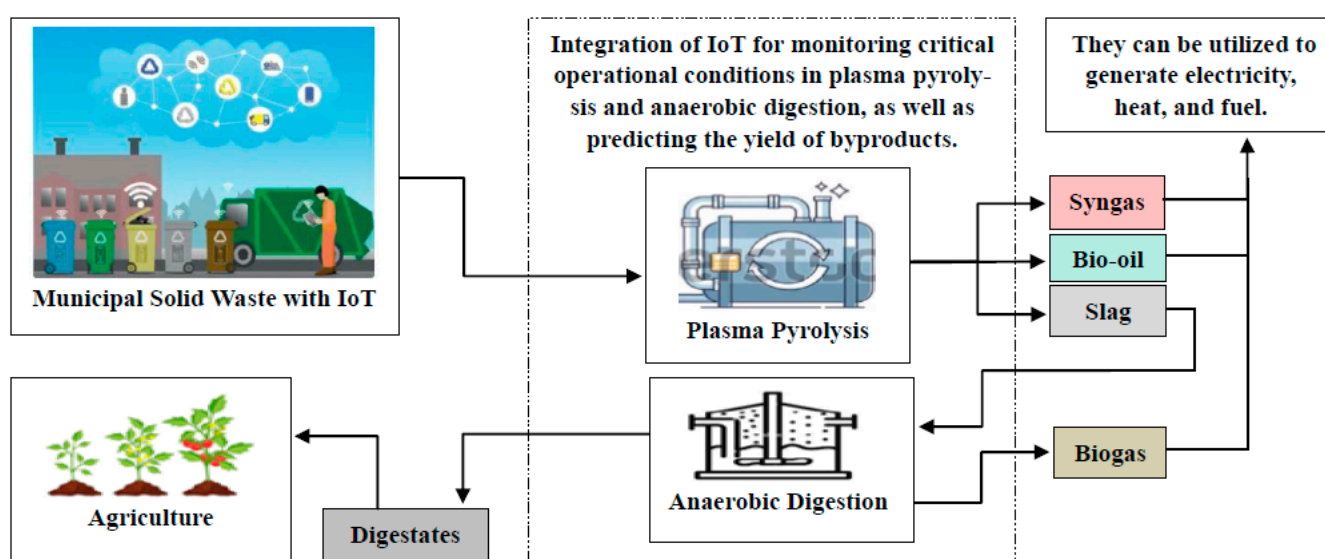


Figure 3. Schematic diagram of municipal solid waste treatment using plasma pyrolysis and co-processing with anaerobic digestion, integrated with the Internet of Things.

The schematic diagram presents an advanced MSW treatment system that integrates plasma pyrolysis and AD, enhanced by the IoT (Figure 3). In this innovative framework, IoT technology facilitates efficient garbage collection [59], directing MSW to the plasma pyrolysis unit [9], which is continuously monitored to maintain optimal operating conditions. This monitoring system aids in the accurate prediction of byproducts, including syngas, bio-oil, and slag. [63,64]. Importantly, plasma pyrolysis not only addresses energy recovery but also ensures the immobilization of HMs and the destruction of toxic and hazardous chemical compounds that may be present in MSW [10,34]. Additionally, the slag from plasma pyrolysis is subsequently channeled into AD as an additive, significantly enhancing biogas production and generating a nutrient-rich digestate. For example, Feng et al. [12] investigated the impact of slag as an additive in the AD of cow manure, as well as the fertility benefits of the resulting digestate. Their findings indicate that all types of steel slag sourced from various steel and iron manufacturers (designated as slag-1, slag-2, and slag-3) significantly enhance the AD process by improving chemical oxygen demand (COD) degradation rate, methane yield, and biogas yield. Specifically, slag-2 exhibited cumulative values COD degradation rate of 58.62%, 274.70 mL/g volatile solids for methane yield, and 507.29 mL/g volatile solids for biogas yield. In addition, AD performance can be optimized by IoT. According to Peter

et al. [65] research, a genetic algorithm (GA) model utilizing an ANN was developed to forecast biogas yields and enhance production processes. The integration of sensors and IoT technologies enabled real-time monitoring of variables influencing biogas output, resulting in an outstanding efficiency rate of 78.2%, with an R^2 of 0.85.

Moreover, Ilangovan et al. [66] introduced a promising model for the online monitoring of the AD process that operates autonomously without the need for human oversight. They evaluated the performance of the biogas digester using a novel algorithm known as the Enhanced Methane Prediction Algorithm (EMPA). The authors compare the EMPA's effectiveness in forecasting methane production and the time required for dataset processing against well-known machine learning methods, including ANN, support vector machines, and Naive Bayes. Notably, the R^2 of the proposed model was 94.84%, which shows high accuracy in predicting methane yields while processing the dataset in a shorter timeframe. In addition, the syngas and bio-oil produced serve as valuable resources for generating electricity, heat energy, and fuel [58], as well as the biogas from AD [67]. Additionally, the digestate produced is an effective biofertilizer that improves soil properties, promoting agricultural productivity. Due to the digestate-containing steel slag possesses excellent thermal stability, making it a promising candidate for use in fertilizers that provide essential nutrients such as nitrogen, phosphorus, and potassium [12]. Therefore, this comprehensive approach not only optimizes MSW management but also aligns with sustainable energy practices, demonstrating the potential for circular economy principles in waste treatment.

6. Techno-Economic of Proposed Technology

The techno-economic analysis of the proposed technology presents a transformative approach to waste treatment and resource recovery. By utilizing plasma pyrolysis, MSW is thermally decomposed at high temperatures, resulting in syngas and slag, both of which can be harnessed for energy production. The incorporation of IoT technologies facilitates real-time monitoring and optimization of the waste processing system, enhancing operational efficiency and reducing costs. Furthermore, the slag generated from plasma pyrolysis can be effectively redirected into AD systems, where it serves as a substrate for biogas production. This dual approach not only maximizes energy recovery from waste but also minimizes landfill dependency, contributing to a circular economy. The synergy between plasma pyrolysis and AD, coupled with smart technology, positions this integrated waste-management strategy as a sustainable solution for urban centers facing escalating waste challenges.

As studied by Li et al. [30], a preliminary economic and environmental assessment of plasma pyrolysis plants reveals several key insights. The primary revenue streams for these facilities stem from tipping fees and the added value of reusable byproducts and end products. A notable challenge of the plasma process is its high electricity consumption, which can make operations costly. However, when the energy content of the produced materials is recycled and utilized, the plasma process can achieve energy self-sufficiency, leading to a reduction in operational costs [10]. For instance, a plasma pyrolysis plant processing 600 tons per day (t/d) of MSW requires a total investment of approximately USD 99.3 million. Revenue is generated from four main sources: electricity, glassy slag, foam glass, and subsidies. The plant is projected to produce 350 kWh of electricity per ton of waste, with an assumed selling price of 9.7 cents per kWh, resulting in annual electricity sales of approximately USD 6.76 million. The glassy slag, estimated at 90,000 tons per year, can be sold for USD 105 per ton, generating around USD 9.45 million. Additionally, foam glass, produced at 10,000 tons per year, is valued at USD 1080.80 per ton for use as thermal insulation, yielding about USD 10.81 million. Together, these two products account for roughly 70% of the total profits. Moreover, considering local subsidy policies, the plant could receive around USD 9.26 per ton, translating to an additional USD 1.85 million annually. Therefore, the

total revenue for the plant would amount to approximately USD 28.87 million per year, significantly higher than the revenue generated by conventional incineration processes, which typically does not exceed USD 9.5 million. This indicates that the overall economic returns from the plasma process are approximately three times greater than those from incineration, highlighting its clear advantages in waste management.

Moreover, Galaly et al. [58] have reported the economic feasibility of treating medical waste under plasma pyrolysis. In a study that focused on energy recovery from medical waste in the Middle East, particularly in the Kingdom of Saudi Arabia, the authors examined the thermodynamic properties of air plasma torches. They investigated flow rates of air between 10 mg/s and 30 mg/s and plasma jet temperatures ranging from 1500 °C to 5000 °C. Key parameters analyzed included power loss, enthalpy, plasma flux, and variations in torch efficiency based on plasma input power and airflow rates. The electrothermal efficiency of the plasma torch was found to range from 42% to 80%, showing an increase with higher input power and gas flow rates. In Makkah, hospitals have a total of 10,500 beds, generating an average annual medical waste of approximately 2,835,000 tons. The pyrolysis process is expected to yield around 2,268,000 tons of pyrolysis oil, translating to an energy equivalent of 90 billion megajoules. The proposed plasma treatment initiative anticipates that after the distillation process, diesel oil production could reach up to 1,928,000 tons, with potential electricity sales estimated at 21 million megawatt-hours. In 2022, the revenue from pyrolysis oil sales amounted to approximately USD 34.44 million, highlighting the economic viability of this waste-to-energy approach.

Furthermore, for Galaly et al. [68], the annual assessment of landfill decomposition hazards in Makkah indicated a significant increase from 168,000 Mg to 297,000 Mg from 1994 to 2022. Correspondingly, pollution control costs rose from USD 4 million to USD 7 million during the same period. Additionally, emissions from plastic waste in the landfill escalated from 128,000 Mg in 1994 to 227,000 Mg by 2022. The thermal plasmas utilized were generated by air plasma torches, crucial components of plasma reactors. Research into the operational characteristics of these torches revealed a data flow rate between 10 and 30 g/s, with current-voltage profiles, applied power, and flow rates contributing to average exit plasma jet temperatures ranging from 15,000 to 19,000 K. The exit jet's average velocity was measured between 1677.3 and 2763.2 m/s, with emitted radiation wavelengths from 153 to 193 nm and beam lengths between 400 and 450 mm. The author found that the recommended plasma method is considered the most environmentally friendly approach for plastic waste processing, achieving a recovery of 317,000 tons of pyrolysis oil in 2022, equating to an energy output of 12.55 billion MJ and an efficiency of 81%. An environmental and economic roadmap was outlined, showing an 80% return on investment, a gross profit margin of 129%, and a payback period of 1.2 years.

This review also suggests incorporating slag into AD to improve biogas production and facilitate the breakdown of organic materials [12]. Zhang et al. [67] conducted an economic analysis of biochar's application in industrial biogas plants, using Singapore as a case study. In 2018, Singapore generated approximately 763,100 tons of food waste (FW), of which only 16.54% was recycled, leaving 83.46% for disposal. Assuming that all FW produced in Singapore has the same volatile solids content of 21.28 wt.% as that used in their study and treated with similar efficiency, the potential annual methane yield could reach 101 million cubic meters. In contrast, a pilot-scale digester without biochar would yield only 73.9 million cubic meters per year. This additional methane yield of 27.1 million cubic meters annually could significantly boost electricity production, offering substantial economic benefits if biochar is available at a reasonable cost. Moreover, the actual prices of slag can vary significantly based on factors such as location, type of slag, and market demand. Actual prices range from about USD 0.00025 per kg for steel slags in areas where

natural aggregates are abundant to nearly USD 0.09 per kg for some ground granulated blast furnace slag [69], unlike biochar, which cost around USD 0.08 to 0.12 per kg [70]. To manage the annual FW of 763,100 tons using biochar-amended AD, the total cost for biochar would be approximately USD 4.06 to USD 6.09 million. However, the net profit from the increased methane yields of 36.67% per year could be around USD 9.99 million, assuming the methane, with a calorific value of about 17.8 kWh/kg, is converted into electricity at a 35% efficiency and sold to the national grid at USD 90.27 per MWh. Therefore, since slag is more affordable compared to the price of biochar, using slag as an additive could be a more economical strategy to enhance the thermophilic AD process and bio-energy production. Consequently, future research should focus on assessing the specific financial potential of the proposed technology.

7. Challenges, Future Perspectives, and Conclusions

The treatment of MSW using plasma pyrolysis and its co-processing with AD presents several challenges and prospects, particularly when integrated with the IoT. One significant hurdle is the complexity of merging data from various sources [71]. Developing algorithms capable of correlating the performance metrics of both systems is essential for optimizing the overall process. Moreover, Plasma pyrolysis operates under extreme conditions, such as high temperatures and corrosive gases, while AD involves variable biological processes that can impact the stability of sensors and devices [64,66]. It is crucial to ensure that IoT components can function reliably in these harsh environments without losing their accuracy or lifespan. Security concerns also arise with the integration of IoT into plasma pyrolysis and AD systems. The interconnected nature of IoT devices increases vulnerability to cyber threats, including data breaches and unauthorized access [71]. Protecting sensitive operational data and maintaining the integrity of both systems is paramount. Furthermore, MSW consists of a wide range of materials, including hazardous substances [4], necessitating a thorough analysis to identify which waste composites are suitable for plasma pyrolysis. This is important because certain waste materials, such as polystyrene, which contains aromatic groups, can adversely affect the yield of byproducts [31].

In conclusion, the treatment of MSW with plasma pyrolysis integrated with IoT technologies offers a transformative approach to resource recovery and energy generation. Plasma pyrolysis effectively converts organic waste into valuable energy byproducts, including syngas and bio-oil, which can be harnessed for the production of fuel, electricity, and heat. Furthermore, the slag or biochar generated from this process can be strategically utilized in AD, enhancing biogas production while promoting the degradation of organic materials. The biogas produced can similarly be converted into energy, thereby contributing to a sustainable energy portfolio. Additionally, the digestate from AD emerges as a nutrient-rich organic fertilizer, which could provide an eco-friendly solution for soil amendment and enhancing agricultural productivity. By leveraging the IoT for real-time monitoring and optimization, plasma pyrolysis can achieve greater operational efficiency, reducing costs and environmental impacts. Collectively, these strategies not only minimize MSW landfill dependency and greenhouse gas emissions but also align with circular bio-economy principles, fostering a sustainable and resilient urban waste-management framework. The holistic approach proposed in this review underscores the potential for a synergistic relationship between waste management, energy production, and agricultural sustainability, paving the way for a greener future.

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